CORE PURE / VECTORS, MATRICES & LINEAR TRANSFORMATIONS

# Reflections in Planes

Questions by topic 7 questions · 84 marks

## Questions


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|---|---|---|
| Question 1 | *9 marks* | 2 |
| Question 2 | *12 marks* | 3 |
| Question 3 | *11 marks* | 4 |
| Question 4 | *11 marks* | 5 |
| Question 5 | *16 marks* | 6 |
| Question 6 | *14 marks* | 7 |
| Question 7 | *11 marks* | 8 |

**1.** A plane passes through the points $C(1, -1, 2)$, $D(3, 0, 1)$ and $E(2, 2, 2)$.

The point $A$ has coordinates $(4, -2, 7)$ and the point $B$ is the reflection of $A$ in the plane.

Find the coordinates of the point $B$. **[9]**

**2.** A line $\ell$ has equation

$$\mathbf{r} = \begin{pmatrix} 4 \\ 2 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

Another line $m$ has equation

$$\mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$$

A plane $\Pi$ contains $m$ and is parallel to $\ell$.

**(i)** Find an equation of $\Pi$, giving your answer in the form $\mathbf{r} \cdot \mathbf{n} = p$. **[4]**

**(ii)** Find the distance between $\ell$ and $\Pi$. **[2]**

**(iii)** Find an equation of the line which is the reflection of $\ell$ in $\Pi$, giving your answer in the form $\mathbf{r} = \mathbf{a} + t\mathbf{b}$. **[6]**

**3.** The line $\ell_1$ has vector equation

$$\mathbf{r} = \begin{pmatrix} 1 \\ 4 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$$

The plane $\Pi_1$ passes through the point $(2, 0, 0)$ and has normal vector

$$\begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$$

**(a)** Find the point of intersection of $\ell_1$ and $\Pi_1$. **[2]**

The line $\ell_2$ is the reflection of the line $\ell_1$ in the plane $\Pi_1$.

**(b)** Show that a vector equation for the line $\ell_2$ is

$$\mathbf{r} = \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$$

where $\mu$ is a scalar parameter. **[7]**

The plane $\Pi_2$ contains the line $\ell_1$ and the line $\ell_2$.

**(c)** Determine a vector equation for the line of intersection of $\Pi_1$ and $\Pi_2$. **[2]**

**4.** The triangle $ABC$ lies in the plane $\Pi$. The vertices are

$$A(1,0,0),\ B(2,2,1) \text{ and } C(1,1,c)$$

where $c$ is a constant.

**(a)** Find, in terms of $c$, $\overrightarrow{BA} \times \overrightarrow{BC}$. **[2]**

Given that $\overrightarrow{BA} \times \overrightarrow{BC} = d\mathbf{i} + 2\mathbf{j} - \mathbf{k}$, where $d$ is a constant,

**(b)** find the value of $c$ and show that $d = -3$. **[2]**

**(c)** find an equation of $\Pi$ in the form $\mathbf{r} \cdot \mathbf{n} = p$, where $p$ is a constant. **[2]**

The point $S$ has position vector $3\mathbf{i} + \mathbf{j} + 2\mathbf{k}$. The point $S'$ is the image of $S$ under reflection in $\Pi$.

**(d)** Find the position vector of $S'$. **[5]**

**5.** The line $\ell$ has equation

$$\mathbf{r} = (\mathbf{i} + \mathbf{k}) + \lambda(\mathbf{i} + 2\mathbf{j} + 3\mathbf{k})$$

where $\lambda$ is a scalar parameter, and the plane $\Pi$ has equation

$$\mathbf{r} \cdot (2\mathbf{i} - \mathbf{j} + \mathbf{k}) = 9$$

**(a)** Find the coordinates of the point of intersection of $\ell$ and $\Pi$. **[2]**

The perpendicular to $\Pi$ from the point $A\,(0, 4, 1)$ meets $\Pi$ at the point $B$.

**(b)** Verify that the coordinates of $B$ are $(4, 2, 3)$. **[3]**

The point $A\,(0, 4, 1)$ is reflected in the plane $\Pi$ to give the image point $A'$.

**(c)** Find the coordinates of the point $A'$. **[2]**

**(d)** Find an equation for the line obtained by reflecting the line $\ell$ in the plane $\Pi$, giving your answer in the form

$$\mathbf{r} \times \mathbf{a} = \mathbf{b}$$

**[9]**

**6.** The plane $\Pi$ has vector equation

$$\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + s \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

The line $\ell_1$ passes through the point $P(4, 1, 2)$ and is perpendicular to $\Pi$.
The line $\ell_1$ meets $\Pi$ at the point $Q$.

**(a)** Find the coordinates of $Q$. **[6]**

Given that the point $R(3, 1, 0)$ lies on $\Pi$,

**(b)** find a Cartesian equation of the plane containing $P$, $Q$ and $R$. **[4]**

The line $\ell_2$ passes through $P$ and $R$.
The line $\ell_3$ is the reflection of $\ell_2$ in $\Pi$.

**(c)** Find a vector equation of $\ell_3$. **[4]**

**7.** The line $\ell_1$ is the intersection of the planes

$$x + y + z = 1$$

and

$$2x - y + z = 4$$

The plane $\Pi$ has equation

$$x + y + 2z = 3$$

The line $\ell_2$ is the reflection of the line $\ell_1$ in the plane $\Pi$.

Find a vector equation of the line $\ell_2$. **[11]**